

where we update the preference model with new AI feedback in order to keep it on the same distribution as the policy produces. We saw that this was valuable with human feedback, and by using AI feedback we can fully automate the process.

Robustness was also another motivation for using chain-of-thought reasoning in this work – we would eventually like AI systems to reason through the hidden risks of certain behaviors, in order to mitigate increasingly subtle and implicit harms.

6.2 Broader Impacts

As with most methods that can control AI behavior, the ideas discussed in this work have a dual use. As we pass from prompting, to RLHF, to the constitutional methods discussed here, we lower the barrier to training AI models that behave in ways their creators intend. This means that these methods also make it easier to train pernicious systems. The supervised methods we have discussed may be particularly accessible, since they do not require an efficient RL implementation with large language models.

A further issue is that by reducing the *need* for human feedback, our constitutional methods make it easier to train and deploy AI systems that have not been thoroughly tested and *observed* by humans. This could lead developers to deploy models with unforeseen failure modes. On the other hand, our method has the benefit that we may no longer need an army of human red teamers to engage in the rather unsavory work of trying to trick AI systems into generating harmful content.

7 Contribution Statement

Model Pre-training: Model pretraining was led by Nicholas Joseph and Sam McCandlish, with help from Tom Brown and Jared Kaplan, and much of Anthropic’s technical staff contributed to the development of our efficient distributed training infrastructure and the underlying machine learning systems. Core contributors include Tom Henighan, Scott Johnston, Sheer El Showk, Nelson Elhage, and Ben Mann. Scott Johnston in particular worked on optimizing pretraining for ML efficiency, while Sheer El Showk, Carol Chen, and Jennifer Zhou worked on data.

Reinforcement Learning: The core RL infrastructure was built by Andy Jones and Kamal Ndousse in collaboration with Shauna Kravec and Dawn Drain. Development of the RL infrastructure has been led by Sam McCandlish and Dario Amodei.

Sampling and Evaluation: Efficient sampling efforts were led by Tom Brown, and Tom Conerly carried out major aspects of the design, implementation and support for the system, with help from Zac Hatfield-Dodds. Many members of Anthropic worked on our framework for evaluations, including Saurav Kadavath, Nicholas Schiefer, Nick Joseph, Tom Henighan, Amanda Askell, Jared Kaplan, Andy Jones, Ethan Perez, Scott Johnston, and Sam McCandlish. Saurav in particular developed the systems for efficient composition of sampling, prompting, and evaluation used for SL and RL CAI, which were one of the primary tools used in this project. Jackson Kernion helped support human feedback data collection.

Cluster: Nova DasSarma and Eli Tran-Johnson managed the research cluster our research depended on and maintained its stability, making this research possible. Many others helped with these efforts, including Ben Mann, Tom Henighan, Sam McCandlish, Andy Jones, Zac Hatfield-Dodds, and Tristan Hume.

Research: Jared Kaplan developed the main ideas in discussion with Yuntao Bai, Amanda Askell, and Saurav Kadavath, and Jared carried out some of the initial experiments. Yuntao developed the method further and designed and carried out most of the experiments in this paper. Amanda helped develop the initial experiments, and Sandipan worked on harmlessness scores and automated generation of prompts.

Writing: This paper was drafted by Yuntao Bai and Jared Kaplan. Other members of Anthropic made miscellaneous contributions and suggestions throughout the writing process.

Other contributions: The ideas explored in this paper developed in conversations with many of Anthropic’s staff, especially Amanda Askell, Deep Ganguli, Sam Bowman, Ethan Perez, Saurav Kadavath, Dario Amodei, Sam McCandlish, Jackson Kernion, Stan Fort, Chris Olah, and Catherine Olsson.

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A Sample critiques and revisions

We show samples of critique and revision from the constitutional method for a variety of hand-written prompts designed to elicit harmfulness. The original response, critique and revision are all sampled from the same 52B helpful RLHF model. We find that the critiques often provide inaccurate criticism. Nonetheless, the first revision often removes most harmful content from the original response, while subsequent revisions make only minor improvements.

We sample four sequential critiques and revisions for PALMS, LaMDA, and InstructGPT prompts, which we provide in our repository. An example is shown below: